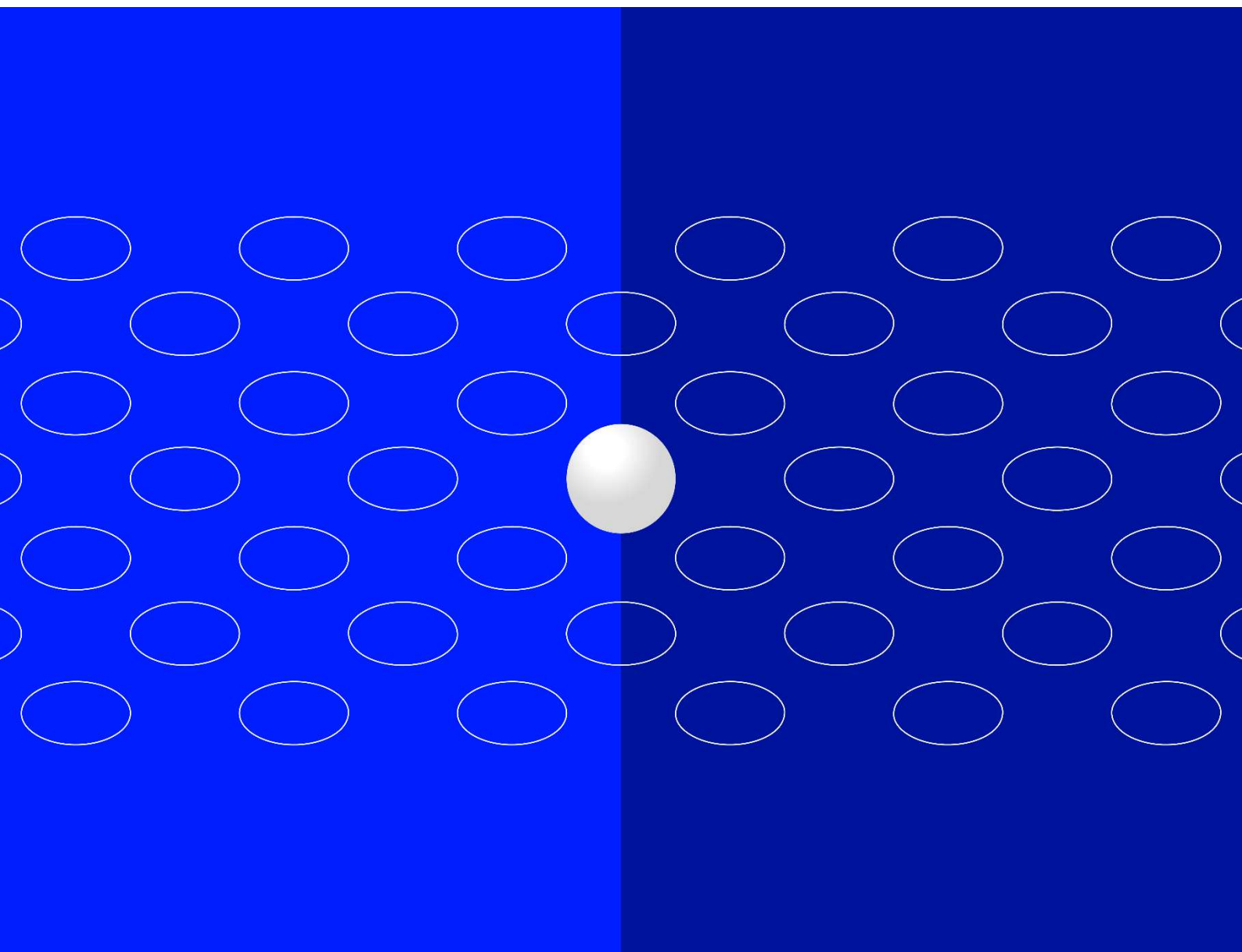


# GROUP TICKER PLANT

GTP 002 – Technical Guide – TRADEcho Pre Trade SI Quote

ISSUE 26.1.2

27 January 2026



LONDON  
STOCK  
EXCHANGE

# Contents

|                                                      |    |
|------------------------------------------------------|----|
| Guide disclaimer .....                               | 4  |
| 1 Documentation.....                                 | 5  |
| 1.1 This guide .....                                 | 5  |
| 1.2 Readership.....                                  | 5  |
| 1.3 Document series .....                            | 5  |
| 1.4 Document history .....                           | 5  |
| 1.5 Enquiries .....                                  | 11 |
| 2. Service description.....                          | 12 |
| 2.1 Reference Data .....                             | 12 |
| 2.2 System Trading Phase .....                       | 13 |
| 2.3 SI Quoting .....                                 | 13 |
| 2.4 Order Book Clear/Product Failure .....           | 13 |
| 2.5 Negative Prices .....                            | 13 |
| 3 Connectivity .....                                 | 14 |
| 3.1 Transmission Standards .....                     | 14 |
| 3.1.1 Multicast Channels .....                       | 14 |
| 3.1.2 Recovery and Replay Services .....             | 14 |
| 3.2 Client Identification.....                       | 14 |
| 4 Message formats .....                              | 15 |
| 4.1 Packet composition .....                         | 15 |
| 4.2 Message types .....                              | 15 |
| 4.3 Sequence Numbers .....                           | 15 |
| 4.4 Timestamps.....                                  | 15 |
| 4.5 Data types .....                                 | 16 |
| 4.6 Message overview – Administrative Messages.....  | 17 |
| 4.7 Message overview – Application Messages.....     | 18 |
| 4.8 Unit header .....                                | 18 |
| 4.9 Administrative Messages (Client-Initiated) ..... | 19 |
| 4.9.1 Login Request .....                            | 19 |
| 4.9.2 Replay Request.....                            | 20 |
| 4.9.3 Recovery Request.....                          | 21 |
| 4.10 Administrative Messages (Server-Initiated)..... | 23 |
| 4.10.1 Heartbeat.....                                | 23 |
| 4.10.2 Login response .....                          | 23 |
| 4.10.3 Replay Response .....                         | 24 |
| 4.10.4 Recovery Response .....                       | 25 |
| 4.10.5 Replay and Recovery Complete .....            | 26 |

---

|        |                                                                          |    |
|--------|--------------------------------------------------------------------------|----|
| 4.11   | Application Messages .....                                               | 27 |
| 4.11.1 | System Event .....                                                       | 27 |
| 4.11.2 | Instrument Directory .....                                               | 28 |
| 4.11.3 | Instrument Status .....                                                  | 30 |
| 4.11.4 | Delete Order .....                                                       | 31 |
| 4.11.5 | Order Book Clear .....                                                   | 32 |
| 4.11.6 | Systematic Internaliser (SI) Quotes .....                                | 33 |
| 5      | Client Data Recovery .....                                               | 35 |
| 5.1    | Replay Channel .....                                                     | 35 |
| 5.1.1  | Establishing a connection .....                                          | 35 |
| 5.1.2  | Sending a request .....                                                  | 35 |
| 5.1.3  | Terminating a connection .....                                           | 35 |
| 5.2    | Recovery Channel .....                                                   | 36 |
| 5.2.1  | Establishing a connection .....                                          | 36 |
| 5.2.2  | Sending a request for an Instrument Level Order Book snapshot .....      | 36 |
| 5.2.3  | Sending a request for a Group or Channel Level Order Book snapshot ..... | 36 |
| 5.2.4  | Sending a request for the Instrument Directory (reference data) .....    | 37 |
| 5.2.5  | Sending a request for an Instrument Status .....                         | 37 |
| 5.2.6  | Terminating a connection .....                                           | 37 |
| 6      | Additional field values .....                                            | 38 |
| 6.1    | Supported currencies .....                                               | 38 |
| 7      | Data Mapping .....                                                       | 39 |
| 7.1    | Conversion of negative values in price fields .....                      | 39 |
| 7.2    | Encoding negative values in price fields .....                           | 39 |
| 7.3    | Decoding negative values in price fields .....                           | 39 |

## Guide disclaimer

The content of this publication is not offered as advice on any particular matter and must not be treated as a substitute for specific advice. In particular, information in this publication does not constitute professional, financial or investment advice and must not be used as a basis for making investment decisions and is in no way intended, directly or indirectly, as an attempt to market or sell any type of financial instrument. LSEG, London Stock Exchange Group plc. and its group undertakings ("London Stock Exchange Group") attempts to ensure that all information used in this publication is accurate at the time of publication. However, all information is provided "as is" and on an "as available" basis and may not be accurate or up to date and the London Stock Exchange Group shall not be liable for decisions made in reliance on it. The information provided by London Stock Exchange Group may or may not have been prepared by London Stock Exchange Group but is made available without responsibility on the part of London Stock Exchange Group. No responsibility is accepted by or on behalf of London Stock Exchange Group for any errors, omissions, or inaccurate information. London Stock Exchange Group does not make any representations and disclaims all express, implied and statutory warranties of any kind, including warranties as to accuracy, timeliness, completeness, merchantability or fitness for any particular purpose. Except as is otherwise indicated, London Stock Exchange Group is the owner of the copyright in all the information and of all related intellectual property rights, including but not limited to all database rights, trademarks, service marks and logos.

Whilst this document may be updated at any time, London Stock Exchange Group will endeavor to provide notice to customers of changes being made. This notice, however, cannot always be guaranteed. The information in this publication is therefore for guidance only.

For further information, please contact your usual London Stock Exchange Group contact.



**LONDON  
STOCK  
EXCHANGE**

# 1 Documentation

## 1.1 This guide

The purpose of this document is to provide a detailed guide to the various message types and message formats as employed by the interface of the Group Ticker Plant.

This guide will be updated and reissued when appropriate to do so.

## 1.2 Readership

This document is particularly relevant to trading, market data and technical teams within member firms, information vendors and other market participants who currently receive or are interested in receiving London Stock Exchange Group market data.

When read in conjunction with other Group Ticker Plant documents, it is intended to provide all required information to develop to, and interact with, our real-time information systems.

## 1.3 Document series

This guide, 'GTP002 – Technical guide', forms part of the documentation library supporting those clients interacting with the Group Ticker Plant. For information, the full series of currently available documentation is outlined below:

- GTP001 – Product Guide
- GTP002 – Technical Guide - Turquoise
- GTP002 – Technical Guide – London Stock Exchange
- **GTP002 – Technical Guide – TRADEcho Pre Trade SI Quote (this document)**
- GTP002 – Technical Guide – TRADEcho MiFID II Post Trade Reporting
- GTP003 – Statistics Guide
- GTP004 – Parameters Guide
- GTP005 – Testing Guide

Further documentation to support activities, such as testing and displaying our data, will be released in due course.

The full library of Group Ticker Plant documentation can be found at:

<https://www.londonstockexchange.com/resources/equities-trading-resources/gtp-technical-specifications>

## 1.4 Document history

This document has been subject to the following iterations:

| Issue     | Date             | Description                                                                                                                                                                                                                                 |
|-----------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19.9.3.1  | 17 January 2019  | 3.11.24 – Added Trade Summary to Turquoise services.                                                                                                                                                                                        |
| 19.9.3.2  | 18 January 2019  | Re-branding Turquoise Lit Auctions™ to Turquoise Plato Lit Auctions™. Turquoise Lit Auctions™ has recently become part of the Plato Co-operation agreement.                                                                                 |
| 19.10     | 26 July 2019     | 3.11.13 – Added Hidden Execution Indicator attribute to Trade message.                                                                                                                                                                      |
| 19.10.2.5 | 20 January 2020  | 6.2, 6.3.5, 7.2, 3.10.7, 3.11.1, 3.11.2, 3.11.13, 3.11.20 – Added the Turquoise NYLON™ Cash Order Book as a source venue for which Market Data will be published by GTP.                                                                    |
| 19.10.3   | 9 August 2019    | 3.11.23 – Updated service breakdown with inclusion of EuroTLX Admin channel.<br>6.5 – Added Security Type ML for Milan Bond Market.<br>5 – Updated section due to decommissioning of Tertiary Feed in GTP Milan.                            |
| 19.10.4   | 23 December 2019 | 3.7 – Off Book Trade message removed. 3.11.3, 3.11.4, 3.11.5, 3.11.6 – Instrument.<br>Directory Messages moved under Application messages.<br>3.11.3, 3.11.5 – Updated Data type of EMS.                                                    |
| 19.11     | 28 February 2020 | 3.11.26 – Trade Summary: added Side of resting orders – message becomes applicable to Borsa and EuroTLX venues<br>4.2.2, 4.2.3 – Updated Recovery Request of Order book.<br>6.2 – Combined Borsa and EuroTLX in a single Source Venue code. |
| 19.11.1   | 16 March 2020    | 4.2.2, 4.2.3 – Minor update to Level 1 Order Book Recovery.                                                                                                                                                                                 |

| Issue   | Date              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19.11.2 | 14 May 2020       | 3.11.14, 3.11.22 – Minor update to footnote and Trade Type definition.<br>3.7 – Updated Intelligent Throttling schema.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 20.2    | 10 August 2020    | <b>Changes for Release 1.4.5.0</b><br>3.11.25, 3.11.26, 3.11.27 – Updated MiFID II Trade, MiFID II Trade Report and MiFID II Trade Cross messages positioning of ILQD, SIZE and LRGS flags.<br>3.11.2 – Updated the datatype of the ADT field from Price to Price4.<br>5, 5.1, 5.2, 5.3, 5.4, 5.5 – Sections rewritten to explain new behaviour and details relating to Failure of Primary Feed and Disaster Recovery using Secondary Data Centre.<br>6.9 – Added the tick sizes for the ATF Market.<br>Removed entries on the revision history prior to 2019.                                                                                                                                                                                                                                                                                                                                                                                          |
| 20.3    | 30 September 2020 | 6.2 – Rebranded Turquoise Global Holdings Limited (TGHL) source venues and added new Turquoise Global Holdings Europe (TGHE) source venues.<br><br>3.11.2, 3.11.3 – Under the Venue Instrument ID field on the Instrument Directory and Instrument Directory – Equities messages added new Instrument Identifiers that will be used by Instruments of the TGHE source venues.<br><br>Note: This change will only go live into the production environment in a Hard Brexit Scenario. If and when this does move to production, there will be a specific notice issued to notify customers.<br><br>3.11.25, 3.11.26 – Added Venue of Publication to distinguish Trade Reports and SI quotes published on EU TRADEcho APA and UK TRADEcho APA.<br><br>Note: This change will only go live into the production environment in a Hard Brexit Scenario. If and when this does move to production, there will be a specific notice issued to notify customers. |
| 20.4    | 10 January 2021   | <b>Changes for Release 1.4.6.0</b><br>All sections – Removed content related to Borsa Italiana and associated venues.<br>3.11.14, 3.11.22 Appended the Trade Qualifier field to the Trade message and the MiFID II Trade message.<br>3.11.22, 3.11.23, 3.11.24 Appended the MMT byte array to the MiFID II Trade, MiFID II Trade Cross and MiFID II Trade Report messages.<br>4.1.3, 4.2.9 Update to how Replay, Recovery and GTP Lite force terminate TCP sessions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 20.5    | 01 February 2021  | L1 Incremental Service is now available for the Turquoise MTF.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 20.6    | 26 March 2021     | 4.1.3, 4.2.9 Removed the update to how Replay, Recovery and GTP Lite force terminate TCP sessions with RST flags, instead FIN will be used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 20.7    | 01 August 2021    | 3.11.27 – Added the Analytics message.<br><br>3.11.1, 3.11.2, 3.11.4 – Updated the diagram to state that the system event message, instrument status message and the instrument directory messages will be published by the Analytics Gateway.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 20.8    | 15 October 2021   | 3.11.15, 3.11.24 – Introduction of BTF trades for Turquoise.<br><br>6.3.2 – 't' will be used for stating the end of Regular Trading on Turquoise Lit™ and Turquoise Plato Lit Auctions™ Order Books.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 20.9    | 25 May 2022       | 3.11.12 – the 2 <sup>nd</sup> bit will be used to indicate Retail LP orders on Turquoise Plato Lit Auctions™ Order Books                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 20.10   | 15 September 2022 | 3.11.12 – the note 'Depth indication won't be provided for Retail LP orders' was deleted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 20.11   | 9 November 2022   | 3.11.23 – Reserved field (Alpha) is converted into 'Total Number of Transactions' (UInt32)<br>4.2.4 – Reject code changed from 'e' to 'a'<br>3.11.17 – Footnote is added to 'Statistic Type' field<br>3.11.17 – 'Not Applicable' value is added to 'Auction Info' field<br>3.11.3 – 'Price Band Tolerances (%)' field converted into 'Reserved Field'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 20.12   | 18 January 2023   | Change made with 20.9 was reverted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 20.13   | 15 February 2023  | Description of rapid login protection is added to section 2.1.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Issue | Date             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21.0  | 20 March 2023    | 3.11.12 – the 2 <sup>nd</sup> bit will be used to indicate Retail LP orders on Turquoise Plato Lit Auctions™ Order Books                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 21.1  | 5 May 2023       | 3.11.17 – Removed 18,19 statistics types since they are not applicable to LSEG Trading Venues                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 22.0  | 5 May 2023       | 3.11.17 – Removed the note that Static and Dynamic Reference Prices are not applicable to LSEG Trading Venues. Description of the functionality is added to GTP 003 - Statistics Guide - Issue 11.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 23.0  | 7 September 2023 | <p>3.11.14 – Typo in 'Trade Qualifier' field is fixed</p> <p>3.11.22 – Typo in 'Instrument Identification Code Type' field is fixed</p> <p>3.11.23 - MiFID II Trade Report (0x54) is updated as follow:</p> <ul style="list-style-type: none"> <li>• New fields are added: Third-country trading venue of execution, Portfolio Transaction Flag, Contingent Transaction Flag, Missing Price</li> <li>• New values are added to the fields: Price Formation Indicator, Reference Price Indicator</li> <li>• Description of the following fields is updated: MiFID Price, Price Currency, Notional Currency, Notional Amount</li> </ul>                                                                                                                                                                                                                      |
| 23.1  | 20 October 2023  | <p>3.11.23 - MiFID II Trade Report (0x54) is updated as follow:</p> <ul style="list-style-type: none"> <li>• Description of field 'MiFID Quantity' is updated</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 24.0  | 21 December 2023 | <p>All sections – Removed content related to London Stock Exchange and Turquoise.</p> <p>Section "5. Product failure" was removed from this document and merged with respective section in GTP001 – "4.0 System overview."</p> <p>The following messages are removed:</p> <ul style="list-style-type: none"> <li>• Instrument Directory – Equities</li> <li>• Add Order – MBO</li> <li>• Add Order Short – MBO</li> <li>• Add Order – MBP</li> <li>• Add Order Short – MBP</li> <li>• Add Order Incremental</li> <li>• Order Modify</li> <li>• Top of Book</li> <li>• Trade</li> <li>• Trade Cross</li> <li>• Statistics Update</li> <li>• FTSE Russell Indices Update</li> <li>• Announcements</li> <li>• Indicative Quote Information</li> <li>• MiFID II Trade</li> <li>• MiFID II Trade Cross</li> <li>• Trade Summary</li> <li>• Analytics</li> </ul> |

| Message              | Updated description of the fields                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System Event         | Event Code                                                                                                                                                 |
| Instrument Directory | Allowed Book Types, Venue Instrument ID, Tick ID, Price Band Tolerances (%), Dynamic Circuit Breaker Tolerances (%), Static Circuit Breaker Tolerances (%) |
| Instrument Status    | Trading Status, Session Change Reason, New End Time, Order Book Type                                                                                       |
| Delete Order         | Order Book Type                                                                                                                                            |
| Order Book Clear     | Order Book Type                                                                                                                                            |
| Statistics           | VWAP, VWAP (on-book only), Turnover, Turnover (on-book only)                                                                                               |

| Issue                 | Date                                                                                                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------|--------------|-------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------|
|                       |                                                                                                                                                                | <div>Statistics Snapshot</div> <div>VWAP, VWAP (on-book only), Turnover, Turnover (on-book only), Official Opening Price, Official Closing Price, Trade High (on-book only), Trade Low (on-book only), Trade High, Trade Low, 52-week Trade High, 52-week Trade Low, Opening Price Indicator, Closing Price Indicator, IAU Price, IAU Paired Size, Imbalance Quantity, Imbalance Direction, Best Closing Bid Price, Best Closing Ask Price, Best Closing Bid Size, Best Closing Ask Size, Trade High Off-Book, Trade Low Off-Book, Auction Type, Last Trade Price, Last Trade Quantity, Last Trade Time, Static Reference Price, Dynamic Reference Price</div>                                                                                                                                                                                                                                                                                                                                                                                                                       |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
|                       |                                                                                                                                                                | <table><tr><th>Message</th><th>Field is renamed</th></tr><tr><td>Instrument Directory</td><td>'Group ID' into 'Segment'<br/>'Underlying ISIN Code' into 'Reserved Field'<br/>'Underlying Instrument ID' into 'Reserved Field'<br/>'Flags' into 'Reserved Field'</td></tr><tr><td>Statistics Snapshot</td><td>'Open Interest' into 'Reserved Field'<br/>'Volatility' into 'Reserved Field'</td></tr><tr><td>Order Delete</td><td>'Previous Yield' into 'Transaction Time' (N/A for TRADEcho)</td></tr><tr><td>MiFID II Trade Report</td><td>'Volume Omission Flag' (offset 329) into 'Volume Omission For Sovereign Debt Flag'</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                      | Message | Field is renamed | Instrument Directory | 'Group ID' into 'Segment'<br>'Underlying ISIN Code' into 'Reserved Field'<br>'Underlying Instrument ID' into 'Reserved Field'<br>'Flags' into 'Reserved Field' | Statistics Snapshot | 'Open Interest' into 'Reserved Field'<br>'Volatility' into 'Reserved Field' | Order Delete | 'Previous Yield' into 'Transaction Time' (N/A for TRADEcho) | MiFID II Trade Report | 'Volume Omission Flag' (offset 329) into 'Volume Omission For Sovereign Debt Flag' |
| Message               | Field is renamed                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
| Instrument Directory  | 'Group ID' into 'Segment'<br>'Underlying ISIN Code' into 'Reserved Field'<br>'Underlying Instrument ID' into 'Reserved Field'<br>'Flags' into 'Reserved Field' |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
| Statistics Snapshot   | 'Open Interest' into 'Reserved Field'<br>'Volatility' into 'Reserved Field'                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
| Order Delete          | 'Previous Yield' into 'Transaction Time' (N/A for TRADEcho)                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
| MiFID II Trade Report | 'Volume Omission Flag' (offset 329) into 'Volume Omission For Sovereign Debt Flag'                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
|                       |                                                                                                                                                                | <div>Changes for Release - TRADEcho 3.17</div> <div>3.5 - Description of data type 'Alpha' is updated</div> <div>3.5 - Description of data type 'UDT' is updated</div> <div>3.11.8 - MiFID II Trade Report (0x54) is updated as follow:</div> <div><ul style="list-style-type: none"><li>New fields are added: 'Market Closing Price Flag', 'NT Large in Scale Flag' and 'NT Pre Trade Transparency Flag'</li><li>New values are added to the fields: Negotiation Indicator, Reference Price Indicator</li><li>The following fields are renamed:<ul style="list-style-type: none"><li>'Price Currency' into 'Price Major Currency (FCA) / Price Currency (ESMA)'</li><li>'Missing Price' into 'Missing Price (ESMA) / Price Conditions (FCA)'</li></ul></li><li>Description of the following fields is updated: MiFID Price, Notional Currency, Negotiation Indicator, Reference Price Indicator, Missing Price (ESMA) / Price Conditions (FCA), Price Major Currency (FCA) / Price Currency (ESMA)</li><li>Added table 'Trade Flags Relevance Across Jurisdictions'</li></ul></div> |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
| 24.1                  | 10 January 2024                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
|                       |                                                                                                                                                                | <div>Changes for Release - TRADEcho 3.17</div> <div>3.11.8 - MiFID II Trade Report (0x54) is updated as follow:</div> <div><ul style="list-style-type: none"><li>'Duplicative Indicator' field is enriched with values related to Intra-Group Trade (IGRP) and Cross-Border Duplicative Trade Report (XBDT)</li><li>'Transaction Category' field is enriched with values related to RFMD Give-Up Trade (GIVE)</li></ul></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
| 24.2                  | 22 January 2024                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
|                       |                                                                                                                                                                | <div>Changes for Release - TRADEcho 3.17</div> <div>Firm Quote book is renamed to SI Quote book in the following messages:</div> <div><ul style="list-style-type: none"><li>3.9.3 - Recovery Request</li><li>3.11.2 – Instrument Directory</li><li>3.11.3 - Instrument Status</li><li>3.11.4 - Order Delete</li><li>3.11.5 - Order Book Clear</li><li>3.11.9 - SI Quotes</li></ul></div> <div>3.11.8 - MiFID II Trade Report (0x54) is updated as follow:</div> <div><ul style="list-style-type: none"><li>'Price Notation' field added support of unset value</li><li>'Secondary Publication' field is renamed to 'Reserved Field'</li></ul></div>                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |
| 24.3                  | 28 March 2024                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |                  |                      |                                                                                                                                                                |                     |                                                                             |              |                                                             |                       |                                                                                    |



| Issue  | Date            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24.4   | 24 April 2024   | 3.5 Data Type section – name of extended ASCII table is corrected<br>3.10.5 Replay and Recovery Complete – typo in field length is corrected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 25.1   | 25 October 2024 | 3.5 Data Type section – Description of MiFID Decimal data type and 'blank' value are updated<br>3.11.8 – MiFID II Trade Report (0x54) <ul style="list-style-type: none"> <li>Precision of MiFID Price, MiFID Quantity, Notional Amount, Quantity in Measurement Units is updated</li> <li>Description of MiFID Quantity is updated</li> <li>Added a new value to reflect combination of flags CLSE and RFPT in the same trades</li> <li>Trade Flags Relevance Across Jurisdictions' table is updated for flags GIVE, IGRP, XBDT in line with change done in version 24.2</li> <li>ORGN flag is removed from Duplicate Flag as it's not applicable to TRADEcho and 'Trade Flags Relevance Across Jurisdictions' table is updated respectively</li> </ul> 3.10.5 – Replay and Recovery Complete <ul style="list-style-type: none"> <li>Added information about possible trading statuses</li> </ul>                                                                                 |
| 25.2   | 12 June 2025    | Typo corrections: <ul style="list-style-type: none"> <li>4.5 Data Types – MiFID Price updated to MiFID Decimal in the last paragraph</li> <li>4.9.2 Repay Request – offset and length of field 'First Message' is updated</li> <li>4.9.3 Recovery Request – sub-header column name changed from 'Hex' to 'Value'</li> <li>4.10.4 Recovery Response – description of field 'Message Type' is updated</li> <li>4.11.1 System Event – missed codes added to field</li> <li>4.11.2 Instrument Directory – description and field name at offset 89 is updated</li> <li>7.3 Base 64 decoding alphabets – column names are updated</li> </ul>                                                                                                                                                                                                                                                                                                                                            |
| 26.1   | 4 August 2025   | <b>SI Quote re-platforming:</b> <ul style="list-style-type: none"> <li>2.0 Service Description section is added</li> <li>2.0 Service Description – 'Instrument Status Management' table moved from 4.1.3. Instrument Status</li> <li>2.0 Service Description – 'Trade Flags Relevance Across Jurisdictions' table moved from 4.11.8 MiFID II Trade Report</li> <li>4.9, 4.10, 4.11 – new SI Quote channels are added</li> <li>4.11.4, 4.11.9 – reserved field is repurposed for Internal ID</li> <li>7.1 Conversion of Order ID - section is removed as no longer relevant</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                             |
| 26.1.1 | 4 August 2025   | <b>GTP002 – Technical Guide – TRADEcho split into:</b> <ul style="list-style-type: none"> <li>GTP002 - Technical Guide – TRADEcho Pre Trade SI Quote (this document)</li> <li>GTP002 - Technical Guide – TRADEcho MiFID II Post Trade Reporting</li> </ul> Information related to MiFID II Trade Reporting service removed: <ul style="list-style-type: none"> <li>2. Service Description</li> <li>4.7 Message overview – Application Messages</li> <li>4.11.8 MiFID II Trade Report</li> <li>4.11.7 Statistics</li> <li>4.11.7 Statistics Snapshot</li> <li>5.2.5 Sending a request for trades</li> <li>5.2.6 Sending a request for statistics snapshot</li> <li>7.1 Conversion of Trade ID</li> <li>Base 64 decoding alphabets</li> </ul> Added a note that Off Book features are not applicable to Pre Trade SI Quote service: <ul style="list-style-type: none"> <li>4.9.3 Recovery Request</li> <li>4.11.2 Instrument Directory</li> <li>4.11.3 Instrument Status</li> </ul> |

---

| Issue   | Date            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                 | <b>Content refresh and clarity Improvement:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 26.1..2 | 27 January 2026 | <ul style="list-style-type: none"><li>• 2.1 Reference Data – mentioned that Currency is also available via Instrument Directory for Non-XLON instruments</li><li>• 4. Message Formats – refreshed wording to provide a clearer, more contemporary description</li><li>• 4.5 Data Types - improved description of default values</li><li>• 4.7 Message overview – Application Messages – update the description of messages and throttling functionality</li><li>• 4.10.5 Replay and Recovery Complete – added clarification that field 'Trading System' may contain default values</li><li>• 5. Client Data Recovery – refreshed wording across subsections, removed outdated content, and removed information not applicable to the trading venue</li><li>• 3.4 Timestamps – reference to regulatory documentation is updated</li><li>• 4.11.2 Instrument Directory – added possible values for the Segment field</li></ul> <p>Please note that all the above modifications are highlighted in <b>amber</b> within the document. It indicates <b>non-material changes</b>, applied to improve clarity and documentation quality.</p> <p>Material changes are shown in <b>red</b> and listed separately.</p> |

---

In subsequent issues of this document, where amendments have been made, these changes will be indicated through the use of **red text**.

---

## 1.5 Enquiries

For further information on Group Ticker Plant, please contact either your Technical Account Manager or the Client Technology Group (UK):

- Telephone: +44 (20)7797 3939
- E-mail: [londontam@lseg.com](mailto:londontam@lseg.com)

Please contact the market data team of LSEG for further information regarding data licensing for this service by e-mailing: [marketdata@lseg.com](mailto:marketdata@lseg.com).

Further information can also be found on our project websites: <https://www.londonstockexchange.com/equities-trading/market-data/group-ticker-plant>

## 2. Service description

### 2.1 Reference Data

#### Start of Day (SOD)

Instrument Directory and Instrument Status messages are disseminated per instrument prior to the System Event – Start of Day (SOD) message.

- The Instrument Directory message (0x70), disseminated on multicast channels, is also available via Recovery.
- Instrument Status messages accompany the Instrument Directory messages, indicating the trading status of each instrument.

**XLON instruments** are published via SI Quote feed. The Instrument Directory includes the following attributes<sup>1</sup>:

- Instrument (GTP Instrument ID) – the only instrument identifier available in other market data messages. This attribute should be used to map market data messages to ISIN.
- ISIN – International Securities Identification Number
- Allowed Book Types – SI Quote Book for SI Quote Feed or Off-Book for MiFID II Post Trade Feed
- Venue Instrument ID – unique identifier across LSE and TRADEcho system
- Segment – classification used by LSE and TRADEcho to group instrument under specific trading model
- Currency – Trading currency of the instrument
- ADT – Average Daily Turnover provided by FCA

Please note that an instrument with the same Venue Instrument ID will be assigned different GTP Instrument IDs on LSE and TRADEcho

**Non-XLON instruments** are also available via Instrument Directory on the SI Quote feed. The Instrument Directory includes:

- Instrument (GTP Instrument ID) – the only instrument identifier available in other market data messages. This attribute should be used to map market data messages to ISIN.
- ISIN – International Securities Identification Number
- Allowed Book Types – SI Quote Book for SI Quote Feed
- Segment – one common code for all Non-XLON instruments
- **Currency – Trading currency of the instrument**

#### Intraday Updates

- SI Quote Feeds: No intraday status changes occur.

#### Instrument Status Management

| TRADEcho SI Quote<br>instrument status | SI Quoting |        | GTP SI Quote Book<br>Instrument status |
|----------------------------------------|------------|--------|----------------------------------------|
|                                        | Cancel     | Accept |                                        |
| Active                                 | Yes        | Yes    | Active                                 |
| Halt                                   | Yes        | Yes    | Reg. Halt                              |
| Instrument suspended                   | Yes        | Yes    | Suspended                              |

<sup>1</sup> A wider list of instrument attributes – including these listed above (except for GTP Instrument ID) – can be found in the instrument reference data file. Full details are provided in MIT401 – Guide to Reference Data Services

---

## 2.2 System Trading Phase

Session transitions are communicated via one System Event message per multicast channel for all allocated instruments.

SI Quote Feeds: System Event message includes:

- Start of Day (SOD) – indicates the beginning of the SI Quoting phase.
- End of Day (EOD) – indicates the closing of the SI Quoting phase. Customers should treat the SI quote book as empty upon receipt. No explicit Order Book Clear or Delete Order messages will be sent.

## 2.3 SI Quoting

Order IDs may repeat within a single trading day after an existing quote is cancelled, if participant resubmits with the same quote identifier. In addition, system may reuse Order ID's across days, and these quotes across days should be treated as separate quotes.

A reserved field is used as Internal ID, which is introduced for system use and participants may ignore this field.

## 2.4 Order Book Clear/Product Failure

- For SI Quote feeds, an Order Book Clear message is published only in case of a failover scenario
- In case of switch, to SDC the system will launch with empty books, so clients must treat every quote active in PDC as expired. The only information that will be republished is instrument data and the start of the day event.

## 2.5 Negative Prices

- SI Quote Feeds: Negative prices are not supported.

---

## 3 Connectivity

### 3.1 Transmission Standards

The Group Ticker Plant employs industry standard data delivery and recovery transmission techniques. Further details are provided below.

Clients should note that network addressing is included as part of GTP004 – Parameters guide.

#### 3.1.1 Multicast Channels

The Group Ticker Plant delivers real-time market data over a number of load-balanced IP multicast channels. Details of the service line allocation across multicast channels can be found in GTP001 – Product guide.

The real-time channels transmit in UDP network packets over IP version 4 (IPv4) Ethernet standards. UDP header information is as defined in the IETF RFC 791 (IPv4) and RFC 768 (UDP) transmission standards. While each UDP network packet will contain a single Unit Header, multiple application messages may be packaged in a single network packet. This is done in an effort to manage client bandwidth requirements. Further details are contained in section 4.3.1 of GTP001 – Product guide.

Clients should subscribe to both primary and secondary market data feeds. While, during normal service, replay and recovery services are only available on the primary market data gateway, through subscription to both primary and secondary feeds clients are able to arbitrage messages – recovering any missed messages on the primary market data feed from the secondary market data feed.

#### 3.1.2 Recovery and Replay Services

The Recovery and Replay unicast channels will guarantee data delivery through use of TCP over IP version 4 (IPv4) Ethernet standards. TCP Header information is defined in the IETF RFC 793 standard and IPv4 is defined in the RFC 791 standard.

### 3.2 Client Identification

All clients are required to request enablement on Group Ticker Plant service lines ahead of access to either CDS or Production environments. Access is permissioned on both the Group Ticker Plant gateways and at our network firewalls.

Upon successful completion of access request, the Group will allocate clients their CompID(s). The CompID should be used by clients to log in to either the replay or recovery services. Each CompID will be permissioned to access all Group Ticker Plant multicast gateways for which a client has the correct commercial agreement(s) in place, each CompID can only be logged in to one replay or recovery service at any one time.

Unlike the existing Millennium Exchange market data solution, the Group Ticker Plant does not require a password to access the replay or recovery services. The removal of password validation was in direct response to client feedback on our existing product provision.

## 4 Message formats

The Group Ticker Plant delivers comprehensive real-time, multi-asset class information through a single, bespoke binary protocol. Developed in close collaboration with clients, the protocol features a clear and intuitive message library that reflects the current structure of our market data model. Its functionality includes:

- Replay and recovery services operate without password validation
- Replay and recovery services function without a logout process
- Statistics messages for the publication of derived information
- An Allowed Book Types field in Instrument Directory messages to identify applicable trading models
- MiFID-compliant market data messages supporting transmission of data in line with regulatory standards

The protocol is designed to be generic and highly extensible, enabling straightforward integration of additional services as the Group Ticker Plant evolves.

### 4.1 Packet composition

The Unit Header is used to deliver all administrative and application messages to and from the server on all multicast and unicast channels. Whilst a Unit Header may contain multiple application messages, it will never contain more than one administrative message. A Unit Header will not contain a combination of both application and administrative messages.

### 4.2 Message types

The Group Ticker Plant broadcasts a library of messages. The messages are categorised as either administrative or application messages. The administrative messages are used on the TCP/IP replay and recovery and GTP Lite services to implement the interaction of a user with the system. The application messages are used to broadcast our real-time data service lines on the multicast channels, and to provide certain static reference data information via replay and recovery services.

Clients should treat each application message as a single standalone instruction, updating their order books and systems appropriately based on the content of the application message. Clients should not program to a multicast stream of messages in an attempt to identify patterns or system logics, as the stream of messages disseminated is subject to change as we optimise our trading and market data technologies. If a client processes each application message in real-time as a standalone instruction, order books will be a true state of the trading engines of the supported markets.

### 4.3 Sequence Numbers

All application messages transmitted by the server on the multicast channels and the replay services are sequenced. The Unit Header only contains the sequence number of the first message; the sequence numbers of any other messages included in the same packet are implied. The sequence number of the next packet can be determined by adding the value in the Message Count field of the Unit Header to the value in the Sequence Number field.

The application messages sent from the server by the recovery service and all administrative messages (including those sent by the client) are un-sequenced.

### 4.4 Timestamps

All Group Ticker Plant application messages contain a timestamp with nanosecond granularity and will be sent in UTC. Timestamps are derived from the supporting infrastructure timestamp, which is synched through various processes including industry standards such as NTP – guaranteeing accuracy to the millisecond – and PTP allowing accuracy to microsecond.

## 4.5 Data types

The fields used within the messages adopt the data types defined below. Where individual field descriptions later in this document reference a value of 'blank' or 'default value', this denotes the default value of the corresponding data type, as specified in the table that follows:

| Data Type            | Length   | Default Value | Description                                                                                                                                                                       |
|----------------------|----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alpha                | Variable | Spaces        | These fields use standard ASCII and Windows-1252 (CP-1252) based on Latin-1 (ISO8859-1) character bytes. They are left justified and padded on the right with spaces.             |
| Bit Field            | 1        | 0 (zero)      | A single byte used to hold up to eight 1-bit flags. Each bit will represent a Boolean flag. The 0 bit is the lowest significant bit and the 7 bit is the highest significant bit. |
| Byte                 | 1        | 0 (zero)      | A single byte used to hold one ASCII character.                                                                                                                                   |
| Date                 | 8        | Spaces        | Date specified in the YYYYMMDD format using ASCII characters.                                                                                                                     |
| Time                 | 6        | Spaces        | Time specified in HHMMSS format using ASCII characters in a 24-hour clock format.                                                                                                 |
| UDT (Unix Date Time) | 8        | 0 (zero)      | Little-Endian encoded 64 bit unsigned integer where; time stamp (in UTC) = (date time per second resolution in unix time format) * 1,000,000,000 + (nanoseconds component).       |
| Price                | 8        | 0 (zero)      | Signed Little-Endian encoded 64 bit integer field with eight implied decimal places.                                                                                              |
| Size                 | 8        | 0 (zero)      | Little-Endian encoded 64 bit unsigned integer with eight implied decimal places.                                                                                                  |
| Price4               | 8        | 0 (zero)      | Signed Little-Endian encoded 64 bit integer field with four implied decimal places.                                                                                               |
| Size4                | 8        | 0 (zero)      | Little-Endian encoded 64 bit unsigned integer with four implied decimal places.                                                                                                   |
| UInt8                | 1        | 0 (zero)      | 8 bit unsigned integer.                                                                                                                                                           |
| UInt16               | 2        | 0 (zero)      | Little-Endian encoded 16 bit unsigned integer.                                                                                                                                    |
| UInt32               | 4        | 0 (zero)      | Little-Endian encoded 32 bit unsigned integer.                                                                                                                                    |
| UInt64               | 8        | 0 (zero)      | Little-Endian encoded 64 bit unsigned integer.                                                                                                                                    |



## 4.6 Message overview – Administrative Messages

| Name                         | Message type |      | Usage                                                                                                          |
|------------------------------|--------------|------|----------------------------------------------------------------------------------------------------------------|
|                              | ASCII        | Hex  |                                                                                                                |
| Heartbeat                    | -            | -    | Used by the server, on the real-time service, to exercise the communication line during periods of inactivity. |
| Login Request                | (soh)        | 0x01 | Used by the client to log in to the replay or recovery channel.                                                |
| Login Response               | (stx)        | 0x02 | Used by the server to accept or reject a login request to the replay or recovery channel.                      |
| Replay Request               | (etx)        | 0x03 | Used by the client to request a retransmission of messages on the replay channel.                              |
| Replay Response              | (eot)        | 0x04 | Used by the server to respond to a retransmission request on the replay channel.                               |
| Recovery Request             | •            | 0x81 | Used by the client to request data on the recovery channel.                                                    |
| Recovery Response            | ,            | 0x82 | Used by the server to respond to a snapshot request on the Snapshot channel.                                   |
| Replay and Recovery Complete | <i>f</i>     | 0x83 | Used by the server to indicate the successful completion of servicing a message replay or a recovery request.  |

## 4.7 Message overview – Application Messages

Each GTP Application Message is listed below with corresponding ASCII and Hex codes. For additional information related to Intelligent Throttling, please refer to GTP Product Guide (GTP001).

| Name                 | Message Type |      | Usage                                                                                                                                                                                                                                       | Intelligent Throttling applied? |
|----------------------|--------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
|                      | ASCII        | Hex  |                                                                                                                                                                                                                                             |                                 |
| System Event         | S            | 0x53 | The message will be broadcast at the <b>start</b> and end of day on TRADEcho. Customers should consider the SI Quote book to be empty following receipt of this message. No explicit Order Book Clear or delete order messages will be sent | N                               |
| Instrument Directory | p            | 0x70 | Used to disseminate a limited set of data for all configured instrument types on the real-time channels                                                                                                                                     | Y                               |
| Instrument Status    | H            | 0x48 | The message is used to communicate status of an instrument<br>Instrument status messages are disseminated prior to system event – start of day message                                                                                      | N                               |
| Delete Order         | D            | 0x44 | Sent to instruct recipients to delete a <b>quote</b>                                                                                                                                                                                        | N                               |
| Order Book Clear     | y            | 0x79 | Sent to instruct recipients to remove all <b>quotes</b> from the book for the specified instrument                                                                                                                                          | N                               |
| SI Quoting           | G            | 0x47 | Publishing Systematic Internaliser (SI) Quotes                                                                                                                                                                                              | N                               |

## 4.8 Unit header

| Field             | Offset | Length   | Type   | Description                                                                |
|-------------------|--------|----------|--------|----------------------------------------------------------------------------|
| Length            | 0      | 2        | UInt16 | Length of the message block including the header and all payload messages. |
| Message Count     | 2      | 1        | UInt8  | Number of payload messages that will follow the header.                    |
| Market Data Group | 3      | 1        | Byte   | Identity of the market data group the payload messages relate to.          |
| Sequence Number   | 4      | 4        | UInt32 | Sequence number of the first payload message.                              |
| Payload           | 8      | Variable | -      | One or more payload messages.                                              |

4.9 Administrative Messages (Client-Initiated)

4.9.1 Login Request

| Market Data Group |  | Channel Content     |  |  | Level 2 Incremental | Replay | Recovery |
|-------------------|--|---------------------|--|--|---------------------|--------|----------|
| Channel 1/5       |  | TRADEcho SI Quoting |  |  | N                   | Y      | Y        |
| Channel 3/7       |  | TRADEcho SI Quoting |  |  | N                   | Y      | Y        |
| Channel L/l       |  | TRADEcho SI Quoting |  |  | N                   | Y      | Y        |
| Channel M/m       |  | TRADEcho SI Quoting |  |  | N                   | Y      | Y        |
| Channel N/n       |  | TRADEcho SI Quoting |  |  | N                   | Y      | Y        |
| Channel O/o       |  | TRADEcho SI Quoting |  |  | N                   | Y      | Y        |

| Field        | Offset | Length | Type   | Description                             |               |
|--------------|--------|--------|--------|-----------------------------------------|---------------|
| Length       | 0      | 2      | UInt16 | Length of message including this field. |               |
| Message Type | 2      | 1      | Byte   | Hex                                     | Meaning       |
|              |        |        |        | 0x01                                    | Login Request |
| Username     | 3      | 8      | Alpha  | CompID assigned to the client.          |               |

### 4.9.2 Replay Request

| Market Data Group | Channel Content     | Level 2 Incremental | Replay | Recovery |
|-------------------|---------------------|---------------------|--------|----------|
| Channel 1/5       | TRADEcho SI Quoting | N                   | Y      | N        |
| Channel 3/7       | TRADEcho SI Quoting | N                   | Y      | N        |
| Channel L/I       | TRADEcho SI Quoting | N                   | Y      | N        |
| Channel M/m       | TRADEcho SI Quoting | N                   | Y      | N        |
| Channel N/n       | TRADEcho SI Quoting | N                   | Y      | N        |
| Channel O/o       | TRADEcho SI Quoting | N                   | Y      | N        |

| Field         | Offset | Length | Type   | Description                                                                                                                               |
|---------------|--------|--------|--------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Length        | 0      | 2      | UInt16 | Length of message including this field.                                                                                                   |
| Message Type  | 2      | 1      | Byte   | <b>Hex</b> <b>Meaning</b>                                                                                                                 |
|               |        |        |        | 0x03                      Replay Request                                                                                                  |
| First Message | 3      | 4      | UInt32 | Sequence number of the first message in range to be retransmitted.                                                                        |
| Count         | 7      | 4      | UInt32 | Number of messages to be resent.                                                                                                          |
| Request ID    | 11     | 4      | UInt32 | The value set in this will be echoed back in the corresponding Replay Response. The system will not validate uniqueness of the set value. |

### 4.9.3 Recovery Request

| Market Data Group | Channel Content     | Level 2 Incremental | Replay | Recovery |
|-------------------|---------------------|---------------------|--------|----------|
| Channel 1/5       | TRADEcho SI Quoting | N                   | N      | Y        |
| Channel 3/7       | TRADEcho SI Quoting | N                   | N      | Y        |
| Channel L/I       | TRADEcho SI Quoting | N                   | N      | Y        |
| Channel M/m       | TRADEcho SI Quoting | N                   | N      | Y        |
| Channel N/n       | TRADEcho SI Quoting | N                   | N      | Y        |
| Channel O/o       | TRADEcho SI Quoting | N                   | N      | Y        |

| Field           | Offset | Length | Type   | Description                                                                                                                                                                                                   |                   |
|-----------------|--------|--------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Length          | 0      | 2      | UInt16 | Length of message including this field.                                                                                                                                                                       |                   |
| Message Type    | 2      | 1      | Byte   | Hex                                                                                                                                                                                                           | Meaning           |
|                 |        |        |        | 0x81                                                                                                                                                                                                          | Recovery Request  |
| Request Level   | 3      | 1      | UInt8  | Defines the level of the request:                                                                                                                                                                             |                   |
|                 |        |        |        | Value                                                                                                                                                                                                         | Meaning           |
|                 |        |        |        | 0                                                                                                                                                                                                             | Instrument        |
|                 |        |        |        | 1                                                                                                                                                                                                             | Group (Segment)   |
|                 |        |        |        | 2                                                                                                                                                                                                             | Multicast Channel |
| Instrument      | 4      | 8      | UInt64 | GTP Instrument identifier if Request Level is 0. Blank if not.                                                                                                                                                |                   |
| Group ID        | 12     | 6      | Alpha  | Group/Segment ID if Request Level is 1. Blank if not.                                                                                                                                                         |                   |
| Order Book Type | 18     | 1      | UInt8  | Only considered if the Request Level is 0. If specified, only data related to the specified order book type is provided. If not specified, data for all available book types for the instrument are provided. |                   |
|                 |        |        |        | Value                                                                                                                                                                                                         | Meaning           |
|                 |        |        |        | 0                                                                                                                                                                                                             | All Books         |
|                 |        |        |        | 1                                                                                                                                                                                                             | SI Quote Book     |
| Source Venue    | 19     | 2      | UInt16 | Venue from which market data is received for the instrument.                                                                                                                                                  |                   |
|                 |        |        |        | Value                                                                                                                                                                                                         | Meaning           |
|                 |        |        |        | 11                                                                                                                                                                                                            | TRADEcho          |

| Field           | Offset | Length | Type   | Description                                                                                                                                                       |
|-----------------|--------|--------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recovery Type   | 21     | 1      | UInt8  | The type of messages to be replayed:                                                                                                                              |
|                 |        |        |        | <b>Value</b> <b>Meaning</b>                                                                                                                                       |
|                 |        |        |        | 0                              Instrument Directory                                                                                                               |
|                 |        |        |        | 1                              Order book                                                                                                                         |
|                 |        |        |        | 2                              All Trades (Not applicable to SI Quote)                                                                                            |
|                 |        |        |        | 3                              Statistics (Not applicable to SI Quote)                                                                                            |
|                 |        |        |        | 4                              Instrument Status                                                                                                                  |
|                 |        |        |        | 5                              Reserved                                                                                                                           |
|                 |        |        |        | 6                              System Event                                                                                                                       |
| Sequence Number | 22     | 4      | UInt32 | Not applicable to SI Quote                                                                                                                                        |
| Request ID      | 26     | 4      | UInt32 | The value set in this will be echoed back in the corresponding Recovery Response and Recovery Complete. The system will not validate uniqueness of the set value. |

## 4.10 Administrative Messages (Server-Initiated)

### 4.10.1 Heartbeat

A Unit Header with a Message Count of zero (0) will be used by the server as the Heartbeat message. Such a message will never increment the sequence number of the real-time multicast channel. The next anticipated sequence number will be included in the Sequence Number to enable recipients to detect gaps on the real-time channel.

### 4.10.2 Login response

| Market Data Group | Channel Content     | Level 2 Incremental | Replay | Recovery |
|-------------------|---------------------|---------------------|--------|----------|
| Channel 1/5       | TRADEcho SI Quoting | N                   | Y      | Y        |
| Channel 3/7       | TRADEcho SI Quoting | N                   | Y      | Y        |
| Channel L/l       | TRADEcho SI Quoting | N                   | Y      | Y        |
| Channel M/m       | TRADEcho SI Quoting | N                   | Y      | Y        |
| Channel N/n       | TRADEcho SI Quoting | N                   | Y      | Y        |
| Channel O/o       | TRADEcho SI Quoting | N                   | Y      | Y        |

| Field        | Offset | Length | Type   | Description                              |                                   |
|--------------|--------|--------|--------|------------------------------------------|-----------------------------------|
| Length       | 0      | 2      | UInt16 | Length of message, including this field. |                                   |
| Message Type | 2      | 1      | Byte   | Hex                                      | Meaning                           |
|              |        |        |        | 0x02                                     | Login Response                    |
| Status       | 3      | 1      | Byte   | Status of the login request.             |                                   |
|              |        |        |        | Value                                    | Meaning                           |
|              |        |        |        | A                                        | Login Accepted                    |
|              |        |        |        | a                                        | CompID Inactive/Suspended         |
|              |        |        |        | b                                        | Login Limit Reached               |
|              |        |        |        | c                                        | Service Unavailable               |
|              |        |        |        | d                                        | Maximum Connections Limit Reached |
|              |        |        |        | e                                        | Failed (other)                    |
|              |        |        |        | f                                        | Invalid CompID or IP Address      |

### 4.10.3 Replay Response

| Market Data Group | Channel Content     | Level 2 Incremental | Replay | Recovery |
|-------------------|---------------------|---------------------|--------|----------|
| Channel 1/5       | TRADEcho SI Quoting | N                   | Y      | N        |
| Channel 3/7       | TRADEcho SI Quoting | N                   | Y      | N        |
| Channel L/I       | TRADEcho SI Quoting | N                   | Y      | N        |
| Channel M/m       | TRADEcho SI Quoting | N                   | Y      | N        |
| Channel N/n       | TRADEcho SI Quoting | N                   | Y      | N        |
| Channel O/o       | TRADEcho SI Quoting | N                   | Y      | N        |

| Field         | Offset | Length | Type   | Description                                                                                                                       |                          |
|---------------|--------|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Length        | 0      | 2      | UInt16 | Length of message including this field.                                                                                           |                          |
| Message Type  | 2      | 1      | Byte   | Hex                                                                                                                               | Meaning                  |
|               |        |        |        | 0x04                                                                                                                              | Replay Response          |
| First Message | 3      | 4      | UInt32 | Sequence number of the first message in range to be retransmitted. This will be zero if Status is not "A".                        |                          |
| Count         | 7      | 4      | UInt32 | Number of messages to be resent, not including any Replay and Recovery Complete messages. This will be zero if Status is not "A". |                          |
| Status        | 11     | 1      | Byte   | Value                                                                                                                             | Meaning                  |
|               |        |        |        | A                                                                                                                                 | Request Accepted         |
|               |        |        |        | D                                                                                                                                 | Request Limit Reached    |
|               |        |        |        | O                                                                                                                                 | Out of Range             |
|               |        |        |        | U                                                                                                                                 | Replay Unavailable       |
|               |        |        |        | c                                                                                                                                 | Concurrent Limit Reached |
|               |        |        |        | e                                                                                                                                 | Failed (Other)           |
| Request ID    | 12     | 4      | UInt32 | Will include the value set as Request ID in the Replay Request message.                                                           |                          |



#### 4.10.4 Recovery Response

| Market Data Group | Channel Content     | Level 2 Incremental | Replay | Recovery |
|-------------------|---------------------|---------------------|--------|----------|
| Channel 1/5       | TRADEcho SI Quoting | N                   | N      | Y        |
| Channel 3/7       | TRADEcho SI Quoting | N                   | N      | Y        |
| Channel L/I       | TRADEcho SI Quoting | N                   | N      | Y        |
| Channel M/m       | TRADEcho SI Quoting | N                   | N      | Y        |
| Channel N/n       | TRADEcho SI Quoting | N                   | N      | Y        |
| Channel O/o       | TRADEcho SI Quoting | N                   | N      | Y        |

| Field           | Offset | Length | Type   | Description                                                                                                                                                                           |                                        |
|-----------------|--------|--------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Length          | 0      | 2      | UInt16 | Length of message including this field.                                                                                                                                               |                                        |
| Message Type    | 2      | 1      | Byte   | Hex                                                                                                                                                                                   | Meaning                                |
|                 |        |        |        | 0x82                                                                                                                                                                                  | Recovery Response                      |
| Sequence Number | 3      | 4      | UInt32 | Sequence number on the real-time channel with which the snapshot is synchronised when Recovery Type = 1, 2, 3, 4. This will be zero for other Recovery Types or if Status is not "A". |                                        |
| Count           | 7      | 4      | UInt32 | Number of messages to follow, not including any Replay and Recovery Complete messages. This will be zero if Status is not "A".                                                        |                                        |
| Status          | 11     | 1      | Byte   | Value                                                                                                                                                                                 | Meaning                                |
|                 |        |        |        | A                                                                                                                                                                                     | Request Accepted                       |
|                 |        |        |        | O                                                                                                                                                                                     | Out of Range                           |
|                 |        |        |        | a                                                                                                                                                                                     | Invalid Group or Instrument            |
|                 |        |        |        | b                                                                                                                                                                                     | Request Limit Reached                  |
|                 |        |        |        | c                                                                                                                                                                                     | Concurrent Limit Reached               |
|                 |        |        |        | d                                                                                                                                                                                     | Invalid Recovery Type or Request Level |
|                 |        |        |        | e                                                                                                                                                                                     | Failed (Other)                         |
| Request ID      | 12     | 4      | UInt32 | Will include the value set as Request ID in the Recovery Request message.                                                                                                             |                                        |

#### 4.10.5 Replay and Recovery Complete

| Market Data Group | Channel Content     | Level 2 Incremental | Replay | Recovery |
|-------------------|---------------------|---------------------|--------|----------|
| Channel 1/5       | TRADEcho SI Quoting | N                   | Y      | Y        |
| Channel 3/7       | TRADEcho SI Quoting | N                   | Y      | Y        |
| Channel L/I       | TRADEcho SI Quoting | N                   | Y      | Y        |
| Channel M/m       | TRADEcho SI Quoting | N                   | Y      | Y        |
| Channel N/n       | TRADEcho SI Quoting | N                   | Y      | Y        |
| Channel O/o       | TRADEcho SI Quoting | N                   | Y      | Y        |

| Field                   | Offset | Length | Type   | Description                                                                                                                                               |                               |
|-------------------------|--------|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Length                  | 0      | 2      | UInt16 | Length of message including this field.                                                                                                                   |                               |
| Message Type            | 2      | 1      | Byte   | <b>Hex</b> <b>Meaning</b>                                                                                                                                 |                               |
|                         |        |        |        | 0x83                                                                                                                                                      | Replay and Recovery Complete  |
| Request ID              | 3      | 4      | UInt32 | Will include the value set as Request ID in the Recovery Request message.                                                                                 |                               |
| Trading Status          | 7      | 1      | Byte   | Current Trading status of the Instrument. Populated only when the message is sent at the end of individual order book snapshots during a trading session. |                               |
|                         |        |        |        | <b>Value</b> <b>Meaning</b>                                                                                                                               |                               |
|                         |        |        |        | 1                                                                                                                                                         | Inactive/Underlying Suspended |
|                         |        |        |        | 2                                                                                                                                                         | Suspended                     |
|                         |        |        |        | 3                                                                                                                                                         | Active                        |
|                         |        |        |        | P                                                                                                                                                         | Halt – Regulatory Halt        |
| Blank if not applicable |        |        |        |                                                                                                                                                           |                               |

## 4.11 Application Messages

### 4.11.1 System Event

| Market Data Group | Channel Content     | Level 2 Incremental | Replay | Recovery |
|-------------------|---------------------|---------------------|--------|----------|
| Channel 1/5       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel 3/7       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel L/I       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel M/m       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel N/n       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel O/o       | TRADEcho SI Quoting | Y                   | Y      | Y        |

| Field        | Offset | Length | Type   | Description                                                  |                    |
|--------------|--------|--------|--------|--------------------------------------------------------------|--------------------|
| Length       | 0      | 2      | UInt16 | Length of message including this field.                      |                    |
| Message Type | 2      | 1      | Byte   | Hex                                                          | Meaning            |
|              |        |        |        | 0x53                                                         | System Event       |
| Timestamp    | 3      | 8      | UDT    | Time the message was generated.                              |                    |
| Event Code   | 11     | 1      | Byte   | Value                                                        | Meaning            |
|              |        |        |        | O                                                            | Start of Day       |
|              |        |        |        | T                                                            | Start of Open      |
|              |        |        |        | P                                                            | Start of Pre Close |
|              |        |        |        | C                                                            | End of Day         |
| Source Venue | 12     | 2      | UInt16 | Venue from which market data is received for the instrument. |                    |
|              |        |        |        | Value                                                        | Meaning            |
|              |        |        |        | 11                                                           | TRADEcho           |

#### 4.11.2 Instrument Directory

| Market Data Group | Channel Content     | Level 2 Incremental | Replay | Recovery |
|-------------------|---------------------|---------------------|--------|----------|
| Channel 1/5       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel 3/7       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel L/I       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel M/m       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel N/n       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel O/o       | TRADEcho SI Quoting | Y                   | Y      | Y        |

| Field                                  | Offset | Length | Type      | Description                                                                                                                                                |
|----------------------------------------|--------|--------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Length                                 | 0      | 2      | UInt16    | Length of message including this field.                                                                                                                    |
| Message Type                           | 2      | 1      | Byte      | <b>Hex</b> <b>Meaning</b>                                                                                                                                  |
|                                        |        |        |           | 0x70Instrument Directory                                                                                                                                   |
| Timestamp                              | 3      | 8      | UDT       | Time the message was generated.                                                                                                                            |
| Instrument                             | 11     | 8      | UInt64    | GTP Instrument identifier.                                                                                                                                 |
| ISIN                                   | 19     | 12     | Alpha     | ISIN Code of an instrument.                                                                                                                                |
| Allowed Book Types                     | 31     | 1      | Bit Field | Defines the order-book types that are allowed for the instrument. Each designated bit represents a book type. '0' means not allowed and '1' means allowed: |
|                                        |        |        |           | <b>Bit</b> <b>Name</b>                                                                                                                                     |
|                                        |        |        |           | 1SI Quote Book                                                                                                                                             |
| Source Venue                           | 32     | 2      | UInt16    | Venue from which market data is received for the instrument.                                                                                               |
|                                        |        |        |           | <b>Value</b> <b>Meaning</b>                                                                                                                                |
|                                        |        |        |           | 11TRADEcho                                                                                                                                                 |
| Venue Instrument ID                    | 34     | 11     | Alpha     | Instrument identifier used by the source venue.                                                                                                            |
| Tick ID                                | 45     | 2      | Alpha     | Not Applicable to TRADEcho                                                                                                                                 |
| Price Band Tolerances (%)              | 47     | 8      | Price     | Not Applicable to TRADEcho                                                                                                                                 |
| Dynamic Circuit Breaker Tolerances (%) | 55     | 8      | Price     | Not Applicable to TRADEcho                                                                                                                                 |
| Static Circuit Breaker Tolerances (%)  | 63     | 8      | Price     | Not Applicable to TRADEcho                                                                                                                                 |

| Field                        | Offset | Length | Type      | Description                                                                                                                         |
|------------------------------|--------|--------|-----------|-------------------------------------------------------------------------------------------------------------------------------------|
| Segment                      | 71     | 6      | Alpha     | Segment the instrument is assigned to:                                                                                              |
|                              |        |        |           | <b>Value</b> <b>Meaning</b>                                                                                                         |
|                              |        |        |           | UVIN                          Non-XLON Instrument Universe                                                                          |
|                              |        |        |           | {SEGM}                      XLON Instrument Universe <sup>3</sup>                                                                   |
| Reserved Field               | 77     | 12     | Alpha     | Reserved for future use.                                                                                                            |
| Security Exchange            | 89     | 11     | Alpha     | Not Applicable to TRADEcho                                                                                                          |
| Currency                     | 100    | 3      | Alpha     | Currency Code as per ISO 4217.<br>For additional currencies supported refer to the Additional Field Values section of this document |
| Reserved Field               | 103    | 1      | Byte      | Reserved for future use.                                                                                                            |
| Reserved Field               | 104    | 4      | Alpha     | Reserved for future use.                                                                                                            |
| Average Daily Turnover (ADT) | 108    | 8      | Price4    | Average Daily Turnover as reported by the Source Venue.                                                                             |
| Reserved Field               | 116    | 8      | Alpha     | Reserved for future use.                                                                                                            |
| Reserved Field               | 124    | 1      | Bit Field | Reserved for future use.                                                                                                            |
| Reserved Field               | 125    | 8      | Price     | Reserved for future use.                                                                                                            |
| Reserved Field               | 133    | 8      | Price     | Reserved for future use.                                                                                                            |

<sup>3</sup> To obtain the full set of instrument attributes, including the segment please refer to "MIT401 – Guide to Reference Data Services for London Stock Exchange venue"

### 4.11.3 Instrument Status

| Market Data Group |  | Channel Content     |  |  | Level 2 Incremental | Replay | Recovery |
|-------------------|--|---------------------|--|--|---------------------|--------|----------|
| Channel 1/5       |  | TRADEcho SI Quoting |  |  | Y                   | Y      | Y        |
| Channel 3/7       |  | TRADEcho SI Quoting |  |  | Y                   | Y      | Y        |
| Channel L/l       |  | TRADEcho SI Quoting |  |  | Y                   | Y      | Y        |
| Channel M/m       |  | TRADEcho SI Quoting |  |  | Y                   | Y      | Y        |
| Channel N/n       |  | TRADEcho SI Quoting |  |  | Y                   | Y      | Y        |
| Channel O/o       |  | TRADEcho SI Quoting |  |  | Y                   | Y      | Y        |

| Field                 | Offset | Length | Type   | Description                                                  |                        |
|-----------------------|--------|--------|--------|--------------------------------------------------------------|------------------------|
| Length                | 0      | 2      | UInt16 | Length of message including this field.                      |                        |
| Message Type          | 2      | 1      | Byte   | Hex                                                          | Meaning                |
|                       |        |        |        | 0x48                                                         | Instrument Status      |
| Timestamp             | 3      | 8      | UDT    | Time the message was generated.                              |                        |
| Instrument            | 11     | 8      | UInt64 | GTP Instrument identifier.                                   |                        |
| Source venue          | 19     | 2      | UInt16 | Venue from which market data is received for the instrument. |                        |
|                       |        |        |        | Value                                                        | Meaning                |
|                       |        |        |        | 11                                                           | TRADEcho               |
| Trading Status        | 21     | 1      | Byte   | Value                                                        | Meaning                |
|                       |        |        |        | 1                                                            | Inactive               |
|                       |        |        |        | 2                                                            | Suspended              |
|                       |        |        |        | 3                                                            | Active                 |
|                       |        |        |        | P                                                            | Halt – Regulatory Halt |
| Session Change Reason | 22     | 1      | UInt8  | Value                                                        | Meaning                |
|                       |        |        |        | 0                                                            | Scheduled Transition   |
| New End Time          | 23     | 6      | Time   | Not Applicable to TRADEcho                                   |                        |
| Order Book Type       | 29     | 1      | UInt8  | Value                                                        | Meaning                |
|                       |        |        |        | 1                                                            | SI Quote Book          |

#### 4.11.4 Delete Order

| Market Data Group | Channel Content     | Level 2 Incremental | Replay | Recovery |
|-------------------|---------------------|---------------------|--------|----------|
| Channel 1/5       | TRADEcho SI Quoting | Y                   | Y      | N        |
| Channel 3/7       | TRADEcho SI Quoting | Y                   | Y      | N        |
| Channel L/I       | TRADEcho SI Quoting | Y                   | Y      | N        |
| Channel M/m       | TRADEcho SI Quoting | Y                   | Y      | N        |
| Channel N/n       | TRADEcho SI Quoting | Y                   | Y      | N        |
| Channel O/o       | TRADEcho SI Quoting | Y                   | Y      | N        |

| Field             | Offset | Length | Type   | Description                                                           |                |
|-------------------|--------|--------|--------|-----------------------------------------------------------------------|----------------|
| Length            | 0      | 2      | UInt16 | Length of message including this field.                               |                |
| Message Type      | 2      | 1      | Byte   | <b>Hex</b>                                                            | <b>Meaning</b> |
|                   |        |        |        | 0x44                                                                  | Delete Order   |
| Timestamp         | 3      | 8      | UDT    | Time the message was generated                                        |                |
| Order ID          | 11     | 8      | UInt64 | Unique identifier of the order.                                       |                |
| Instrument        | 19     | 8      | UInt64 | GTP Instrument identifier.                                            |                |
| Side              | 27     | 1      | Byte   | <b>Value</b>                                                          | <b>Meaning</b> |
|                   |        |        |        | B                                                                     | Buy Order      |
|                   |        |        |        | S                                                                     | Sell Order     |
| Order Book Type   | 28     | 1      | UInt8  | <b>Value</b>                                                          | <b>Meaning</b> |
|                   |        |        |        | 1                                                                     | SI Quote Book  |
| Source venue      | 29     | 2      | UInt16 | Venue from which market data is received for the instrument.          |                |
|                   |        |        |        | <b>Value</b>                                                          | <b>Meaning</b> |
|                   |        |        |        | 11                                                                    | TRADEcho       |
| Previous Price    | 31     | 8      | Price  | Price of the order that was deleted from the book.                    |                |
| Previous Quantity | 39     | 8      | Size   | Quantity of the order that was deleted from the book.                 |                |
| Internal ID       | 47     | 8      | UInt64 | Internal ID for system use. This field should be ignored by consumers |                |

#### 4.11.5 Order Book Clear

| Market Data Group | Channel Content     | Level 2 Incremental | Replay | Recovery |
|-------------------|---------------------|---------------------|--------|----------|
| Channel 1/5       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel 3/7       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel L/l       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel M/m       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel N/n       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel O/o       | TRADEcho SI Quoting | Y                   | Y      | Y        |

| Field           | Offset | Length | Type   | Description                                                  |                  |
|-----------------|--------|--------|--------|--------------------------------------------------------------|------------------|
| Length          | 0      | 2      | UInt16 | Length of message including this field.                      |                  |
| Message Type    | 2      | 1      | Byte   | <b>Hex</b>                                                   | <b>Meaning</b>   |
|                 |        |        |        | 0x79                                                         | Order Book Clear |
| Timestamp       | 3      | 8      | UDT    | Time the message was generated.                              |                  |
| Source venue    | 11     | 2      | UInt16 | Venue from which market data is received for the instrument. |                  |
|                 |        |        |        | <b>Value</b>                                                 | <b>Meaning</b>   |
|                 |        |        |        | 11                                                           | TRADEcho         |
| Instrument      | 13     | 8      | UInt64 | GTP Instrument identifier.                                   |                  |
| Order Book Type | 21     | 1      | UInt8  | <b>Value</b>                                                 | <b>Meaning</b>   |
|                 |        |        |        | 1                                                            | SI Quote Book    |



#### 4.11.6 Systematic Internaliser (SI) Quotes

| Market Data Group | Channel Content     | Level 2 Incremental | Replay | Recovery |
|-------------------|---------------------|---------------------|--------|----------|
| Channel 1/5       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel 3/7       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel L/l       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel M/m       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel N/n       | TRADEcho SI Quoting | Y                   | Y      | Y        |
| Channel O/o       | TRADEcho SI Quoting | Y                   | Y      | Y        |

| Field                 | Offset | Length | Type   | Description                                                            |                    |
|-----------------------|--------|--------|--------|------------------------------------------------------------------------|--------------------|
| Length                | 0      | 2      | UInt16 | Length of message including this field.                                |                    |
| Message Type          | 2      | 1      | Byte   | Hex                                                                    | Meaning            |
|                       |        |        |        | 0x47                                                                   | SI Quote           |
| Timestamp             | 3      | 8      | UDT    | Time the message was generated.                                        |                    |
| Order ID              | 11     | 8      | UInt64 | Unique identifier of the order.                                        |                    |
| Side                  | 19     | 1      | Byte   | Value                                                                  | Meaning            |
|                       |        |        |        | B                                                                      | Buy Order          |
|                       |        |        |        | S                                                                      | Sell Order         |
| Size                  | 20     | 8      | Size   | Displayed Size of the order.                                           |                    |
| Instrument            | 28     | 8      | UInt64 | GTP Instrument identifier.                                             |                    |
| Price                 | 36     | 8      | Price  | Limit price of the order. Implied price if instrument trades in yield. |                    |
| Internal ID           | 44     | 8      | UInt64 | Internal ID for system use. This field should be ignored by consumers  |                    |
| Source venue          | 52     | 2      | UInt16 | Venue from which market data is received for the instrument.           |                    |
|                       |        |        |        | Value                                                                  | Meaning            |
|                       |        |        |        | 11                                                                     | TRADEcho           |
| Order Book Type       | 54     | 1      | UInt8  | Value                                                                  | Meaning            |
|                       |        |        |        | 1                                                                      | SI Quote Book      |
| Participant           | 55     | 11     | Alpha  | Identity of trading participant that submitted the order.              |                    |
| Order Type            | 66     | 1      | 66     | Value                                                                  | Meaning            |
|                       |        |        |        | 0                                                                      | SI Quote (default) |
| Reserved              | 67     | 10     | Alpha  | Reserved field.                                                        |                    |
| Currency <sup>4</sup> | 77     | 3      | Alpha  | Currency of the SI Quote price.                                        |                    |

<sup>4</sup> For an order on the given currency on the order book, no order priority or price prioritization is done for SI Quotes

| Field                | Offset | Length | Type  | Description                                                                                                                         |
|----------------------|--------|--------|-------|-------------------------------------------------------------------------------------------------------------------------------------|
| Venue of Publication | 80     | 4      | Alpha | Identification of the regulatory regime under which the transaction was published. The value sent by the source venue is passed on. |
|                      |        |        |       | <b>Value</b> <b>Meaning</b>                                                                                                         |
|                      |        |        |       | ECHOPublished under UK Regulation                                                                                                   |
|                      |        |        |       | ECEUPublished under EU Regulation                                                                                                   |

## 5 Client Data Recovery

The Group Ticker Plant provides both replay and recovery services to facilitate data recovery.

Should a gap in sequence numbers be noticed on both the primary and secondary feed, or following a failure at client site, clients should assume that instrument order books are stale and should initiate one of the below data recovery processes to refresh systems.

### 5.1 Replay Channel

The replay service provides clients the ability to request a finite number of application messages as disseminated on the real-time multicast channel. Provisioned to facilitate client recovery following a small scale data loss, the replay service caches the last 65,000 application messages published for each multicast channel.

Clients are permitted to log on to the replay service a finite number of times each day and, following successful login, subsequently submit a finite number of requests each day. Whilst these counters can be reset intraday by the Group, this will be done only in an emergency situation and should not be relied upon as normal practice.

Each client CompID is permitted to log on to the Production replay service 3,000 times per day, submitting a maximum of 3,000 requests per day. Clients are permitted to queue a maximum of 10 requests at any one time. Should any of these parameters be reached, the replay server will respond with an explanatory reason code.

#### 5.1.1 Establishing a connection

Clients should send a Login Request message to the appropriate target replay service IP gateway address and port. Validation of appropriate credentials as configured on the Group Ticker Plant will be done against both the CompID, as supplied in the message sent by the client, and the incoming source IP address. Upon successful validation, the replay service responds with a Login Response message of Status 'A' – the session should now be considered active. Clients should wait for the server's response prior to submitting replay requests. Any requests submitted prior to authentication will be ignored.

Should the attempted connection fail for any reason, the server will respond with a Login Response message which will contain a reason code. This reason code will facilitate the diagnosis of the failed login request.

#### 5.1.2 Sending a request

Once successful connection is established, clients can queue requests for retransmission of missed messages. The Replay Request message should include the first sequence number of the range of messages to be retransmitted and the total number of messages subsequent to the first missed message required. A Request ID can also be included if required – this is not validated by the replay service.

Should the server accept the request for message retransmission, clients will receive a Replay Response message of Status 'A.' If a Request ID was specified, this will be included in the Replay Response message. This will be immediately followed by the requested message stream. The completion of the request will be marked by the sending of a Replay and Recovery Complete message. This will include the original Request ID if specified but will not include the current Trading Status of the instrument.

Should the replay request fail, the server will respond with a Replay Response message which will contain a reason code. This reason code will facilitate the diagnosis of the failed replay request.

Submitted requests will be processed serially, but the capability of the replay service will be split across any logged in CompID. This may mean that the performance of the replay solution may differ, dependent upon the number of CompIDs logged in to the service at any given time.

#### 5.1.3 Terminating a connection

Clients will not be required to log out from the replay service. Instead, immediately after the completion of the request, the replay server will terminate the connection with the client.

Clients should note that, upon successful login to the replay service, a request should be submitted within five seconds or the server will force-logout the client.

Replay, Recovery and GTP Lite services will close connection to user via message with FIN flag in case termination of connection is initiated by the service itself.

## 5.2 Recovery Channel

The recovery service facilitates a client's resynchronisation with the order book following a large scale data loss for the which the replay service is insufficient. Following successful login, clients are able to request the following:

- A snapshot of the order book for any active instrument in the Market Data Group
- The current trading status of an instrument
- The full set of reference data for an instrument

In a similar approach to that of the replay service, clients are permitted to log in to the recovery service a finite number of times each day and, following successful login, subsequently submit a finite number of requests each day. While these counters can be reset intraday by the Group, this will be done only in an emergency and should not be relied upon as normal practice. Each client CompID is permitted to log on to the Production recovery service 3,000 times per day, submitting a maximum of 3,000 requests per day. Clients are permitted to queue a maximum of 10 requests at any one time. Should any of these parameters be reached, the recovery server will respond with an explanatory reason code.

### 5.2.1 Establishing a connection

Clients should send a Login Request message to the appropriate target recovery service IP gateway address and port. Validation of appropriate credentials as configured on the Group Ticker Plant will be done against both the CompID as supplied in the message by the client and the incoming source IP address. Upon successful validation, the recovery service will respond with a Login Response message of Status 'A' – the session should now be considered active. Clients should wait for the server's response prior to submitting recovery requests. Any requests submitted prior to authentication will be ignored.

Should the attempted connection fail for any reason, the server will respond with a Login Response message which will contain a reason code. This reason code will facilitate the diagnosis of the failed login request.

### 5.2.2 Sending a request for an Instrument Level Order Book snapshot

Following receipt of a Login Response message of Status 'A,' used to confirm successful login to the recovery service, clients may submit a request for a snapshot of the current order book using the Recovery Request message. The Recovery Request message should indicate that the client requires an instrument order book snapshot – the Recovery Type field should be '1,' and the Request Level field should be '0.' Clients are also permitted to request a specific order book, either Electronic or SI Quote, in the Order Book Type field. Clients may also include a Request ID in the Recovery Request message – this is not validated by the server.

The server will transmit a Recovery Response message which should indicate the successful acceptance of the request. The Recovery Response message will also include the Request ID if specified by the client. The Recovery Response message will also include the real-time channel sequence number with which the snapshot is synchronised. Clients should buffer all messages as received on the real-time channel with a sequence number greater than that received in the Recovery Response message.

If active and displayable **quotes** are present on the order book, the server will transmit an Order Book Clear message followed by a message stream to allow the rebuild of the order book. The level 2 incremental services should be rebuilt by processing all SI Quote messages as disseminated following the Recovery Response message. The buy side will always be transmitted first. Following transmission of the order book, the server will disseminate a Replay and Recovery Complete message.

If no active displayable **quotes** are present, the server will disseminate only an Order Book Clear message followed by the Replay and Recovery Complete message. This message includes the real-time trading status of the instrument if the instrument is currently in a trading session. To complete the request, following transmission of the order book for an instrument, the server will disseminate a final Replay and Recovery Complete message. If a Request ID was specified by the client, this will be included.

### 5.2.3 Sending a request for a Group or Channel Level Order Book snapshot

Following receipt of a Login Response message of Status 'A,' used to confirm successful login to the recovery service, clients may submit a request for a snapshot of the current order book using the Recovery Request message. The Recovery Request message should indicate that the client requires Group or Channel level order book snapshots – the Recovery Type field should be '1,' and the Request Level field should be either '1' or '2'. If a Group or Channel level recovery is requested by the client, all order books will be disseminated – the Order Book Type field is not processed by the server. Clients may also include a Request ID in the Recovery Request message. This is not validated by the server.

The server will transmit a Recovery Response message which should indicate the successful acceptance of the request. The Recovery Response message will also include the Request ID, if specified by the client, and the real-time channel sequence number with which the snapshot is synchronised. Clients should buffer all messages as received on the real-time channel with a sequence number greater than that received in the Recovery Response message.

If active and displayable **quotes** are present on the order book, the server will transmit an Order Book Clear message, followed by a message stream to allow the rebuild of the order book.

The level 2 incremental services should be rebuilt by processing all Add Order Incremental messages as disseminated following the Recovery Response message. The buy side will always be transmitted first. The level 1 order book recovery will

publish an Order Book Clear message followed by the current best Bid and Ask aggregated levels in a single TOB message – similar to that of the real-time service.

For instruments with no active displayable quotes present, the server will disseminate only an Order Book Clear message followed by the Replay and Recovery Complete message. This message includes the real-time trading status of the instrument if the instrument is currently in a trading session. To complete the full request, following transmission of all order books, the server will disseminate a final Replay and Recovery Complete message. If a Request ID was specified by the client, this will be included. The final Replay and Recovery Complete message will not contain a value in the Trading Status field.

#### 5.2.4 Sending a request for the Instrument Directory (reference data)

Following receipt of a Login Response message of Status 'A,' used to confirm successful login to the recovery service, clients may submit a request for Instrument Directory messages using the Recovery Request message. The Recovery Request message should indicate that the client requires Instrument Directory messages. The Recovery Type field should be '0.' Clients should also indicate the level of the request – an individual instrument, a segment or for all instruments on the multicast channel. Clients may also include a Request ID in the Recovery Request message. This is not validated by the server.

The server will transmit a Recovery Response message which should indicate the successful acceptance of the request. The Recovery Response message will also include the Request ID, if specified by the client. The Recovery Response message will also include the real-time channel sequence number of the last Instrument Directory message sent. This will be followed by the Instrument Directory message(s) as requested by the client. Successful 'Group' or 'Multicast Channel' requests will result in the dissemination of all configured instruments on the Group Ticker Plant at that request level, irrespective of their trading status. Instrument Directory messages are available via Recovery once they have been published by multicast.

Customers should note that, should a request be made for an instrument not supported by the targeted multicast group, the server will respond with the rejection Status of 'a.'

The completion of the recovery request will be indicated through the dissemination of a Replay and Recovery Complete message. The Trading Status field of the Replay and Recovery Complete message will only be populated if the original request level was 'Instrument.' The Replay and Recovery Complete message will also include the Request ID, if specified by the client.

#### 5.2.5 Sending a request for an Instrument Status

Following receipt of a Login Response message of Status 'A,' used to confirm successful login to the recovery service, clients may submit a request for the current Trading Status of an instrument on any supported market by using the Recovery Request message. The Recovery Request message should indicate that the client requires Instrument Status messages. The Recovery Type field should be '4.' Clients should also indicate the level of the request – an individual instrument, a segment or for all instruments on the multicast channel. Clients may also include a Request ID in the Recovery Request message. This is not validated by the server.

The server will transmit a Recovery Response message which should indicate the successful acceptance of the request. The Recovery Response message will include the Request ID, as specified by the client. This message will also include the real-time multicast channel sequence number of the last real-time message disseminated. Clients should buffer all messages as received on the real-time channel with a sequence number greater than specified. The total number of Instrument Status messages to be disseminated will be indicated in the Count field.

Following dissemination of the Recovery Response message, the server will transmit one or more Instrument Status messages, the number dependent upon the level of the original request. The current Trading Status of an instrument is indicated in the Instrument Status message.

The server will disseminate a Replay and Recovery Complete message to indicate the successful completion of the request. The Trading Status field will only be populated if the request level was 'Instrument'. The Replay and Recovery Complete message will also include the Request ID if specified by the client.

If no Instrument Status messages exist to satisfy the original request, the server will transmit a Recovery Response message followed immediately by a Replay and Recovery Complete message.

#### 5.2.6 Terminating a connection

Clients will not be required to log out from the recovery service. Instead, immediately after the completion of the request, the recovery server will terminate the connection with the client.

Clients should note that, upon successful login to the recovery service, a request should be submitted within five seconds or the server will force-logout the client.

Replay, Recovery and GTP Lite services will close connection to user via message with FIN flag in case termination of connection is initiated by the service itself.

---

## 6 Additional field values

### 6.1 Supported currencies

| Code | Description   |
|------|---------------|
| EUX  | Euro – cents  |
| USX  | Cent          |
| GBX  | GB pennies    |
| ZAC  | 100th of RAND |
| ITL  | Italian LIRA  |

## 7 Data Mapping

### 7.1 Conversion of negative values in price fields

Signed data types, such as Price and Price4, might indicate negative values where applicable. Negative numbers are represented using the "Sign and magnitude" method. Encoding and decoding examples are provided in the following sections.

### 7.2 Encoding negative values in price fields

Decimal value = -1

Decimal value with eight implied decimal places = -100000000

Remove sign bit = 100000000

Convert to binary = 00000000 00000000 00000000 00000000 00000101 11110101 11100001 00000000

Add sign bit = 10000000 00000000 00000000 00000000 00000101 11110101 11100001 00000000

Hex value = 80 00 00 00 05 f5 e1 00

Hex value converted to Little endian = 00 e1 f5 05 00 00 00 80

### 7.3 Decoding negative values in price fields

Received bytes in hex = 00 e1 f5 05 00 00 00 80

Change the byte order to big endian = 80 00 00 00 05 f5 e1 00

Convert to binary = 10000000 00000000 00000000 00000000 00000101 11110101 11100001 00000000

Most significant bit is set. Therefore this is a negative value.

Remove the sign bit = 00000000 00000000 00000000 00000000 00000101 11110101 11100001 00000000

Convert to decimal = 100000000

Add sign to decimal = -100000000

Mark eight implied decimal places -1.00000000